

Andrew Thomas

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Personal Statement

Electronic Engineer and 5G Software Engineer at Accelercomm, specializing in developing cutting-edge physical layer solutions for satellite communication. Proficient in software development, embedded systems, and AI applications, with a keen interest in exploring advanced machine learning frameworks. Achieved MEng in Electrical and Electronic Engineering at Lancaster University, awarded the prestigious UKESF Scholarship, and achieved 3A* and 1A at A-Levels.

Always seeking new challenges and opportunities to further develop breadth of skills in the fields of electronic, embedded and communications engineering.

Technical Skills

- **Programming Languages:** C, C++, Python (PyTorch, NumPy, Pandas)
 - **Embedded Systems:** I2C, SPI, CAN protocols; STM32, Arduino microcontrollers
 - **Software Development:** Linux systems, thread management, CMake, Jenkins, Bash
 - **Circuit Design:** PCB analog/digital design (Altium, KiCad); PCB milling and fabrication
 - **Simulation Tools:** MATLAB, Ansys, COMSOL for thermal-mechanical simulation
 - **Prototyping & Testing:** Wire-bonding, soldering, laser milling; TFT probing and analysis
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Professional Experience

Accelercomm, Southampton – Graduate Software Engineer

October 2023 – Present

- Developed a Google Test framework to validate 3GPP 5G physical channels, generating 5G NR slot scenarios using MATLAB.
- Implemented DPDK-based hardware accelerator drivers, enhancing system efficiency.
- Acquired in-depth knowledge of standards including 3GPP, O-RAN, nFAPI, and eCPRI.

Lancaster Formula Student, Lancaster University – Instrumentation Lead

September 2022 – July 2023

- Led the development of a CAN-based telemetry network, integrating sensors using I2C/SPI with EV systems for real-time data collection.
- Implemented RTOS task management on STM32 microcontrollers, optimizing priority handling.
- Designed many PCBs for the control of the EV and for the onboard telemetry network.

CSA Catapult, Newport – Advanced Packaging Intern

July 2021 – July 2022

- Improved wire bonding and die attachment processes for micro-scale MMIC and VCSEL dies.
- Conducted thermo-mechanical simulations, reducing thermal stress in semiconductor designs.
- Collaborated with industrial partners to prototype innovative semiconductor applications.

Additional Roles

- **Training Captain, Lancaster University Athletics Club (2020–2021):** Designed tailored training plans for diverse skill levels.

- **Volunteer, Morecambe Foodbank & Refugee Community Kitchen** (2019–2020):
Coordinated logistics and service delivery under time-sensitive conditions.
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Education

Lancaster University – MEng Electrical and Electronic Engineering, Sep 2018 – July 2023

Lancaster University - Electrical and Electronic Engineering with 2 Scholarships

Part 1 – 69.2 %

Modules include: Engineering Management, Electrical Circuits and Power Systems, Electromagnetics & RF Engineering, Digital Electronics, Power Electronics and Applications, Integrated Circuit Engineering, Optoelectronics and wireless communications & Digital Signal Processing.

Part 2 – 70.4 %

Modules include: Advanced Embedded Systems, Terahertz Science and Technology, Microengineering & High frequency electronics.

Epsom College

September 2012 – July 2018

- **A-Levels:** Maths (A*), Further Maths (A*), Physics (A*), Chemistry (A)
 - **GCSEs:** 10 subjects including Maths, Physics, and Chemistry (A*/A/B)
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Projects & Interests

- **AI Development:** Built convolutional neural networks for image classification using Kaggle datasets.
- **Hobbyist Projects:** Explored microcontroller communication protocols and designed small-scale PCB systems.
- **Scientific Research Enthusiast:** Regularly follows advancements in physics, cosmology, and AI via sources such as Sabine Hossenfelder and PBS Space Time.
- **Personal Pursuits:** I'm a keen runner both to keep fit and channel my competitive side. ranked in the Top 1000 in the UK (Half Marathon 1hr10, 5k 15:30). Self-taught Classical piano player.